Pilot search analysis

12 November 2026 search | Rechecked 13 November 2026

What I ran

I ran the Python search script locally after the AI environment could not access the database APIs. The pilot ran from 10:25:53 to 10:28:26 UTC. It tested 12 queries on PubMed (6), OpenAlex (5) and Europe PMC (1), saving the raw responses and query metadata. All 12 succeeded.

Results

347 exported records; 296 unique; 51 duplicates removed (14.7%). There are 123 medical and 173 technical unique records. The archived 28 file hashes, exact query strings and counts were checked again. Four groups of similar titles (8 records) still require manual version checking.

Query	Hits	Export	P	I	N	Decision
MED-v1	5273	30	3	15	12	Discard broad control
MED-v2	317	30	15	2	13	Main medical
MED-v3	17	17	0	9	8	Local supplement
MED-v4	82	30	6	21	3	Conditional
TECH-v1	751	30	12	9	9	Discard control
TECH-v2	803	30	12	9	9	Keep
TECH-v3	51841	30	17	5	8	Discard broad control
TECH-v4	599	30	13	7	10	Keep
TECH-v5	2050	30	18	3	9	Keep
TECH-v6	3075	30	19	0	11	Keep
TECH-v7	13188	30	23	1	6	Keep
MED-v5	164	30	19	2	9	Main supplement

P = directly relevant title; I = possible/indirect; N = peripheral. These are AI-assisted title labels, not human inclusion decisions. Database hit counts overlap and must not be added. Each sample contains at most 30 ranked results.

Medical search: what changed

MED-v1 returned 5,273 hits. MED-v2 uses synonyms, MeSH terms, dates and evidence-type filters and returned 317 hits; its directly relevant title count increased from 3/30 to 15/30. That makes screening more manageable, but it does not prove improved recall. MED-v5 added 16 records not present in MED-v2's first 30 (14 shared). MED-v3 found only 17 records, mostly indirect or case reports; a Tunisian affiliation does not prove the study took place in Tunisia. I cannot infer a national prevalence from these results. MED-v4 is useful only if the main reviews leave gaps about treatment or fertility.

Technical search: what changed

TECH-v1 and TECH-v2 shared 28 of their first 30 records. I retain TECH-v2 because it explicitly includes biomedical terms and a time window. TECH-v3 returned 51,841 hits, too broad for this project. TECH-v4 to v7 focus on citations, uncertainty, factuality and multilingual retrieval. OpenAlex is ranked search, not strict Boolean filtering. I will use these targeted searches and follow citations rather than claim an exhaustive AI review.

What this means for my project

The earlier literature notes identify related work on citation checks, refusal, evidence order and claim-level certification. This makes a broad novelty claim unsafe. The contribution should be the controlled comparison of evidence applicability and verification in one domain. I want to test risk at equal coverage, not reward a model for refusing everything.

Limits and next step

The pilot is title-based and ranked, with limited samples; it does not give global search precision or a final PRISMA count. The candidate references and evidence matrix are leads, not included studies. Next: screen titles/abstracts, inspect full texts and verify the actual support and quality of each source.

Reproduction: review/pilot_run_local contains the original run; review/query_analysis.csv and review/record_title_coding.csv contain the derived analysis. Run code/bibliographic_search/verify_pilot.py from the repository root.